

GPE, thus there is no work done on the Moon.

14 Ans: A

$$\frac{1}{2} m \langle c^2 \rangle = \frac{3}{2} kT \rightarrow c_{\text{rms}} = \sqrt{\frac{3kT}{m}}$$

$$\frac{c'_{\text{rms}}}{c_{\text{rms}}} = \sqrt{\frac{T'}{T}} \rightarrow \frac{0.8c_{\text{rms}}}{c_{\text{rms}}} = \sqrt{\frac{T'}{295}}$$

$$T' = 188.8 \text{ K} = -84.4 \text{ }^\circ\text{C}$$

15 Ans: C